📑 Basic Information:

Experiment No: 10

Date:

Title: File Handling in Java to Write, Read, and Count “pe” and “pi” Occurrences.

🎯 Objective /Aim:

To use Java file handling to write text to a file, read its contents, and count how many times the character sequences “pe” and “pi” occur, ignoring case.

🛠️ System Requirements:

Hardware:

-> Standard PC/Laptop

-> Intel Core i3/i5 processor or equivalent

-> Minimum 4 GB RAM

-> Minimum 500 MB free

Software:

->Operating System: Windows / Linux / macOS

-> Java Development Kit (JDK 17 or JDK 21)

-> Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code

-> Command Prompt / Terminal

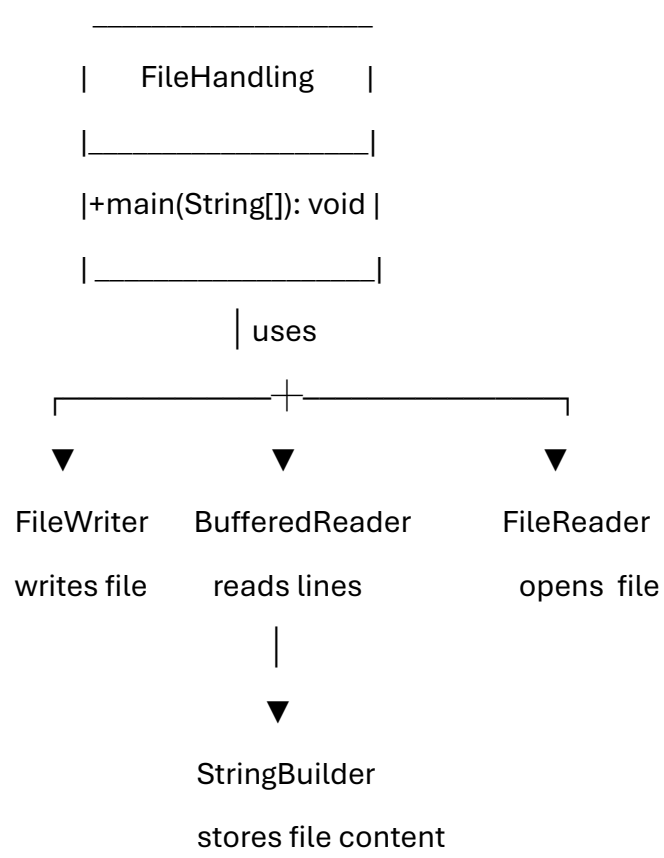
Algorithm/Logic Description:

1. Start the program.
2. Store the given text in a string.
3. Open sample.txt using FileWriter.
4. Write the text to the file and close the writer.
5. Open sample.txt using FileReader and BufferedReader.
6. Read the file line by line and store its contents.
7. Close the reader and convert the contents to lowercase.
8. Set the pe and pi counters to zero.
9. Check each pair of adjacent characters:

- If the pair is pe, increase the pe counter.
- If the pair is pi, increase the pi counter.

10. Display both occurrence counts.
11. Stop the program.

UML Class Diagram:



Source Code:

```
import java.io.FileWriter;
import java.io.FileReader;
import java.io.BufferedReader;
import java.io.IOException;
// Main class for writing to and reading from a file
public class FileHandling {
   public static void main(String[] args) throws IOException {
      // Text to write into the file
      String text = "Peter Piper picked a peck of pickled peppers\n"
```

```
        + "A peck of pickled peppers Peter Piper picked\n"
        + "If Peter Piper picked a peck of pickled peppers\n"
        + "Where's the peck of pickled peppers Peter Piper picked?";
// Write the text into sample.txt
FileWriter fw = new FileWriter("sample.txt");
fw.write(text);
fw.close();
// Open sample.txt for reading
BufferedReader br = new BufferedReader(
    new FileReader("sample.txt"));
String line;
StringBuilder content = new StringBuilder();
// Read the file one line at a time and collect its contents
while ((line = br.readLine()) != null) {
  content.append(line).append("\n");
}
// Close the reader after reading the file
br.close();
// Convert all text to lowercase so counting ignores letter case
String data = content.toString().toLowerCase();
// Counters for occurrences of "pe" and "pi"
int peCount = 0;
int piCount = 0;
// Check each pair of adjacent characters
for (int i = 0; i < data.length() - 1; i++) {
  String pattern = data.substring(i, i + 2);
  // Count the "pe" pattern
  if (pattern.equals("pe")) {
    peCount++;
```

```
        }
        // Count the "pi" pattern
        if (pattern.equals("pi")) {
            piCount++;
        }
    }
    // Display the occurrence counts
    System.out.println("'pe' - no of occurrences - "+ peCount);
System.out.println("'pi' - no of occurrences - " + piCount);
  }
}
```

Relationship:

FileHandling uses FileWriter to write text to sample.txt, then uses FileReader and BufferedReader to read it. It stores and processes the text with StringBuilder and String, then counts occurrences of "pe" and "pi".

Output Screenshot:



Result: The program writes the given text to sample.txt, reads it back, and counts the specified character pairs without distinguishing uppercase and lowercase. It finds 20 occurrences of “pe” and 10 occurrences of “pi.”